

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4 Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.
5. Create a client-server network in Cisco Packet Tracer by configuring a web server

and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weight ag e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understand in g of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technol ogi es	Good understandi ng with proper integration of most concepts	Basic understandi ng ; limited or partial integration	Poor understandi ng ; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementat i on with minor issues	Basic implementa ti on with errors or inefficiencies	Incorrect or incomplete implement ati on
Use of Tools	20	Effective and advanced use of tools/technologi es with proper configuration	Appropria te use of tools with correct execution	Limited or inconsiste nt use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Document ati on	10	Clear, well structured documentation with diagrams, code, and results	Organized documenta tio n with necessary details	Basic documenta tio n with missing details	Poor or incomplete documenta tio n

EXPERIMENT 1

Two PCs Connected Through a Switch and Ethernet Frame Analysis

1. Aim

To design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer, configure IPv4 addresses, verify connectivity using ping, capture the packet exchange using Wireshark, identify Ethernet frame fields, and explain MAC addressing at the Data Link layer.

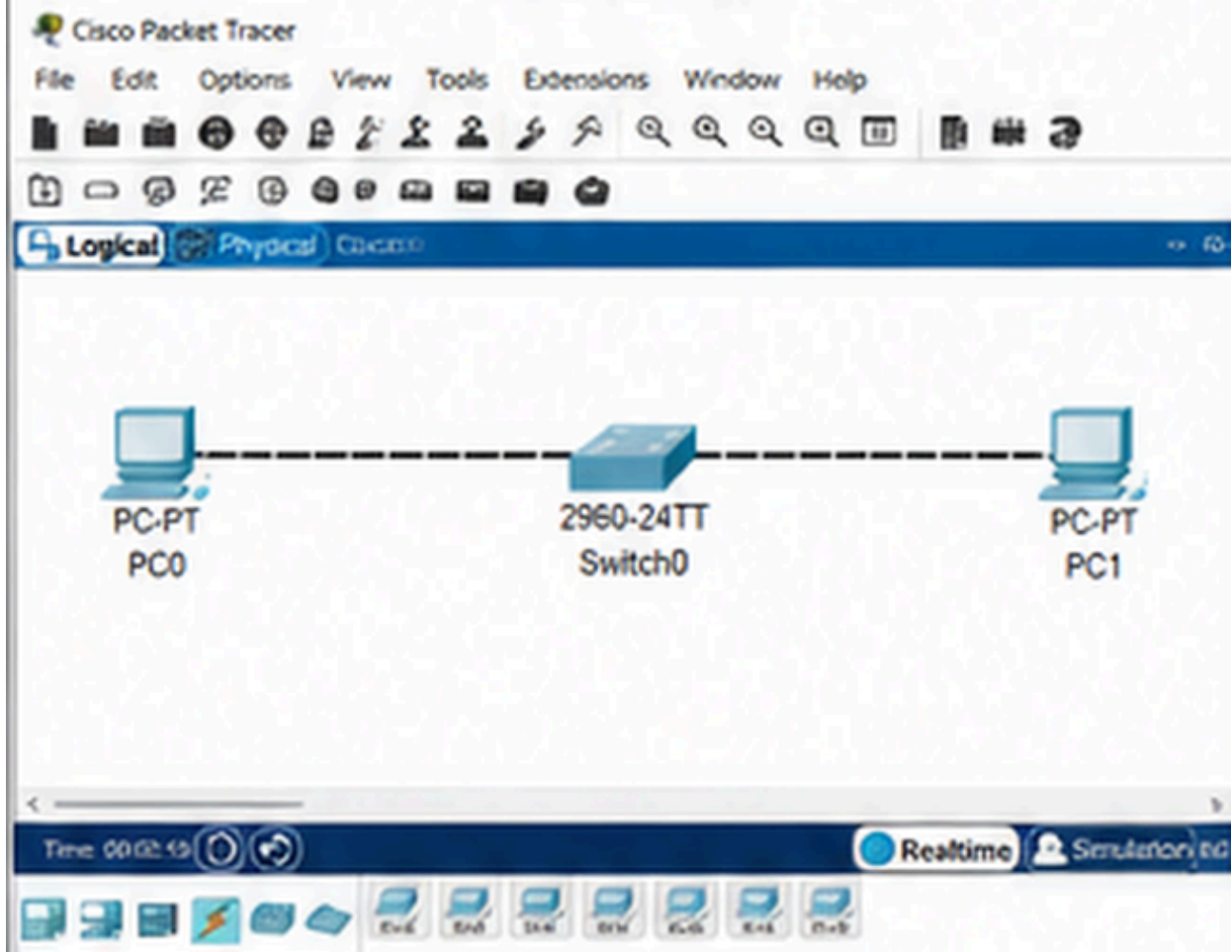
2. Software Required

Cisco Packet Tracer, Wireshark, Windows Command Prompt.

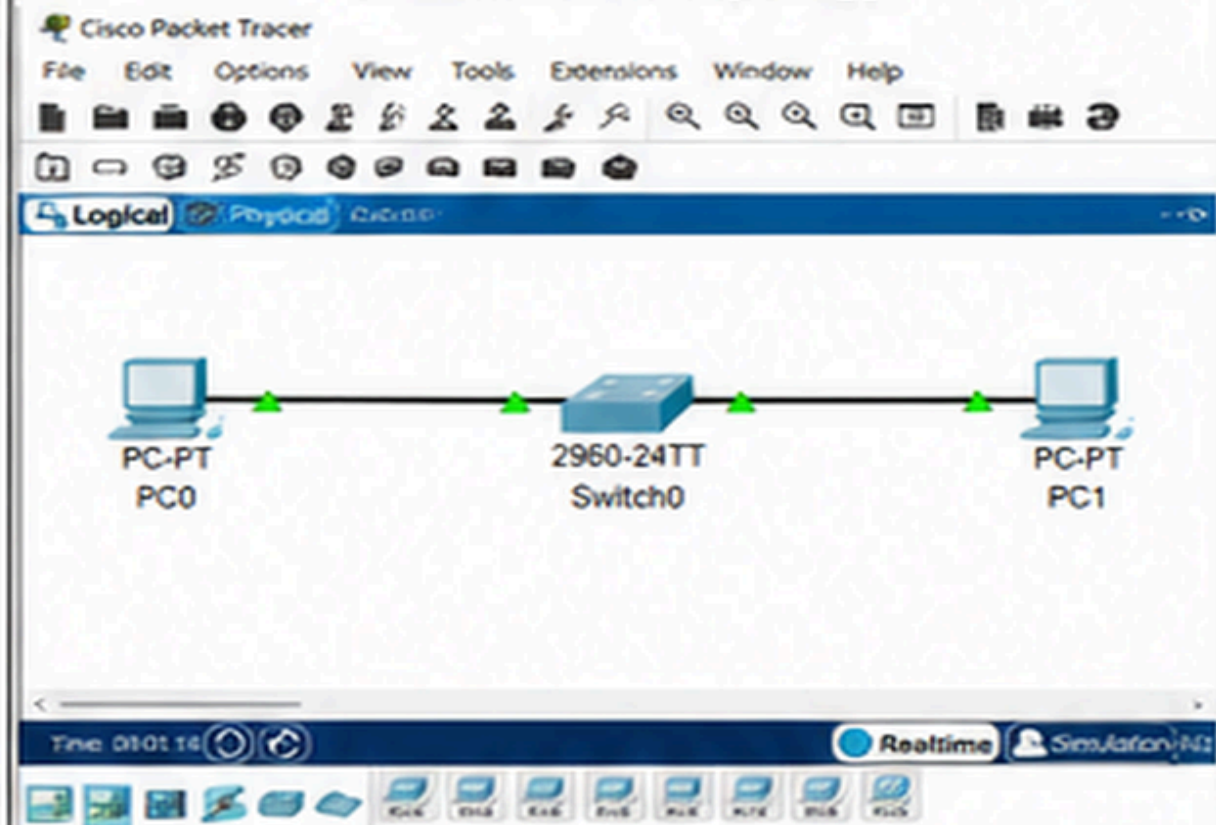
3. Procedure

1. Open Cisco Packet Tracer and place two PCs and one 2960 switch.
2. Connect PC0 and PC1 to the switch using Copper Straight-Through cables.
3. Configure PC0 with IP address 192.168.1.10 and subnet mask 255.255.255.0.
4. Configure PC1 with IP address 192.168.1.20 and subnet mask 255.255.255.0.
5. From PC0 Command Prompt, execute ping 192.168.1.20.
6. Open Wireshark, capture ICMP traffic, and inspect the Ethernet II fields.

1. Topology



2. Connected Topology



3. PC0 IP Configuration

PC0

Physical Config Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.10

Subnet Mask 255.255.255.0

Default Gateway

DNS Server

6. Wireshark – Ethernet II Frame

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Google Chrome

No.	Time	Source	Destination	Protocol	Length	Info
18	0.533170	192.168.1.10	192.168.1.20	ICMP	98	Echo (ping) req
19	0.5340139	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) rep
19	0.5340020	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) rep
19	0.5340009	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) rep
19	0.5340090	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) rep

Frame 18: 98 bytes on wire (784 bits), 98 bytes captured (784 bits)
Ethernet II, Src: 00:E0:F9:66:60:01, Dst: 00:E0:F9:66:68:02
> Destination: 00:E0:F9:66:68:02

5. Ping from PC0 to PC1

PC0

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

6. Result

The two PCs were configured in the same LAN and connectivity was verified using ping. Ethernet frame fields were identified in Wireshark

EXPERIMENT 2

TWO LANs CONNECTED THROUGH A ROUTER AND IPv4 HEADER ANALYSIS

1. Aim

To create two LANs connected through a router, configure IPv4 addresses, subnet masks and default gateways, verify end-to-end communication using ping, capture ICMP packets using Wireshark, analyze IPv4 header fields, and explain Network-layer forwarding.

2. Software Required

Cisco Packet Tracer, Wireshark and Windows Command Prompt.

3. Network Configuration

PC0: 192.168.1.10 / 255.255.255.0 / Gateway 192.168.1.1

Router G0/0: 192.168.1.1 / 255.255.255.0

Router G0/1: 192.168.2.1 / 255.255.255.0

PC1: 192.168.2.10 / 255.255.255.0 / Gateway 192.168.2.1

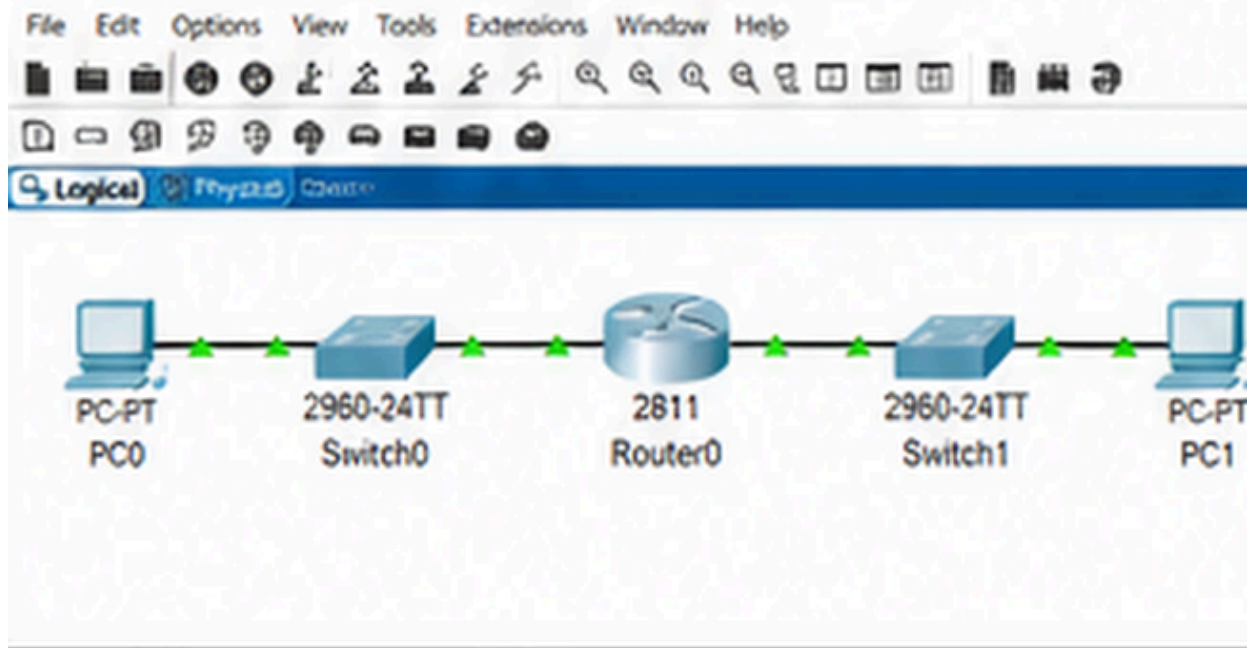
4. Procedure

7. Place PC0, Switch0, Router0, Switch1 and PC1.
8. Connect the two LANs through the router.
9. Configure PC0 and PC1 with the addresses and default gateways shown above.
10. Configure Router0 G0/0 and G0/1 and use no shutdown.
11. Use show ip interface brief to verify the interfaces.
12. From PC0, ping 192.168.2.10 and capture the ICMP packet in Wireshark.

Router CLI commands:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
end
show ip interface brief
```


1. Complete Topology

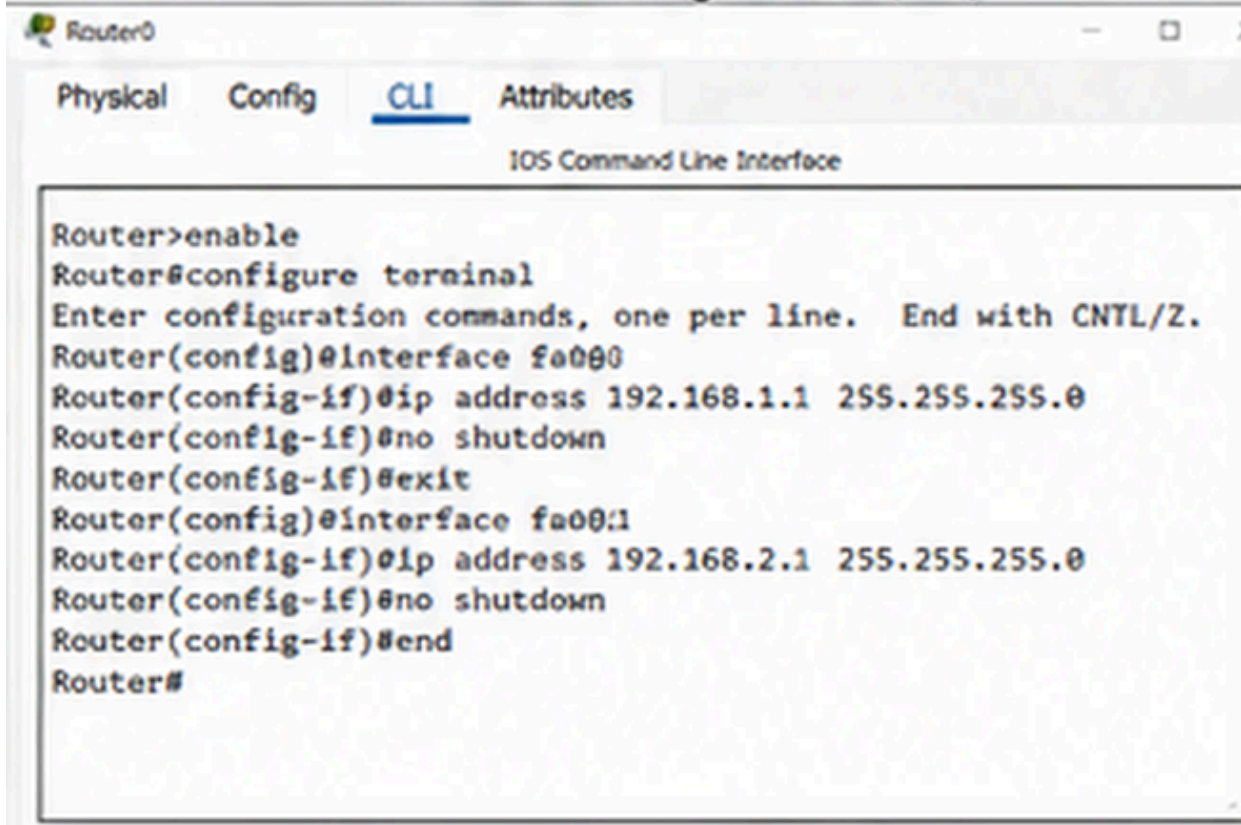


2. PC0 IP Configuration

The screenshot shows the 'PC0' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the following settings:

Field	Value
IP Configuration	Static
IPv4 Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	

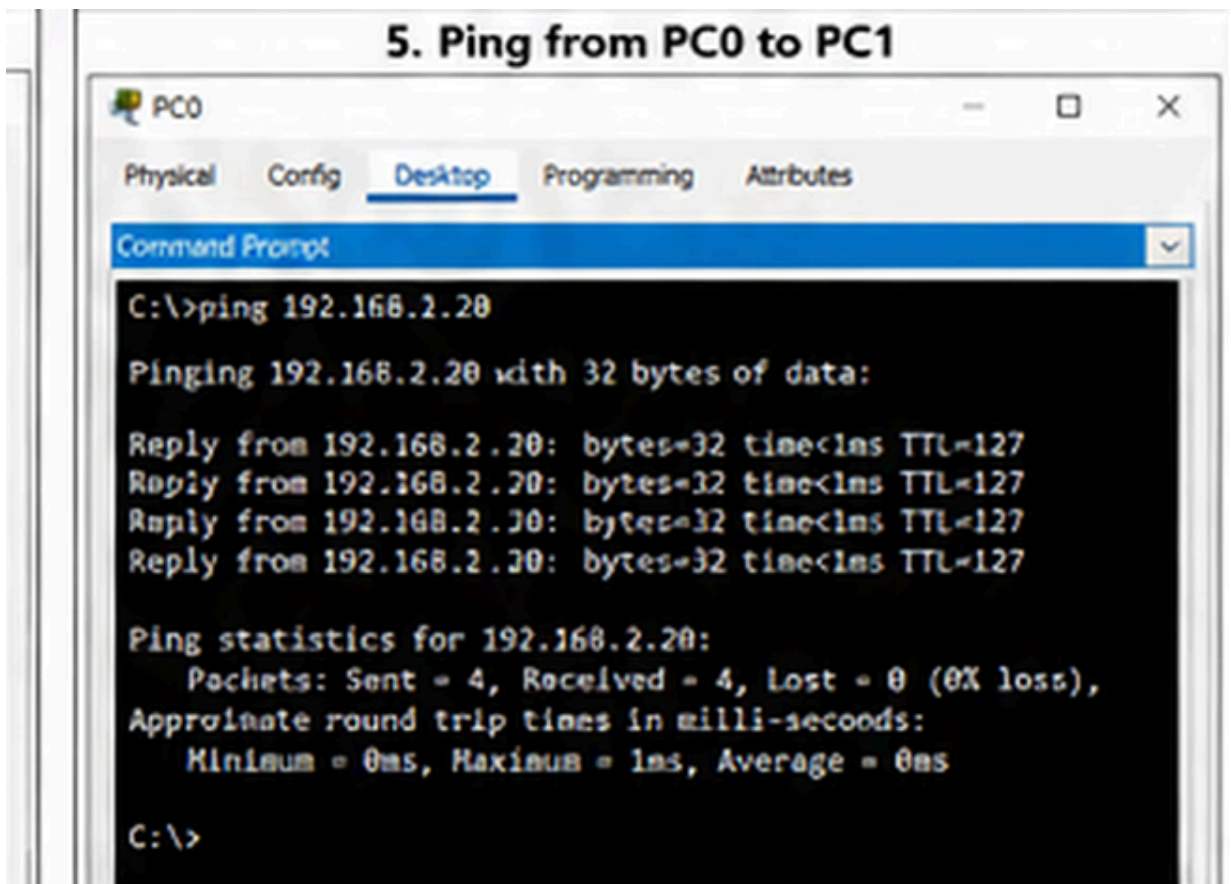
4. Router Configuration (CLI)



The screenshot shows a window titled 'Router0' with tabs for 'Physical', 'Config', 'CLI' (selected), and 'Attributes'. Below the tabs is the text 'IOS Command Line Interface'. The main area displays a series of commands entered in the CLI:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface fa0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
```

5. Ping from PC0 to PC1



The screenshot shows a window titled 'PC0' with tabs for 'Physical', 'Config', 'Desktop' (selected), 'Programming', and 'Attributes'. A 'Command Prompt' window is open, displaying the results of a ping command:

```
C:\>ping 192.168.2.20

Pinging 192.168.2.20 with 32 bytes of data:

Reply from 192.168.2.20: bytes=32 time<1ms TTL=127
Reply from 192.168.2.20: bytes=32 time<1ms TTL=127
Reply from 192.168.2.20: bytes=32 time<1ms TTL=127
Reply from 192.168.2.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Result

Two LANs were configured through a router and end-to-end communication was verified using ping. IPv4 header fields were identified using Wireshark